HOW TO USE THIS TEMPLATE

Delete this page and every purple "Guidance" note before submission

This template is prescribed for the End Semester Examination (Individual Project) of CB23F35 – Low-Code Application Development using Zoho Creator (2023 Regulations). Each student designs, builds and deploys one complete business application on Zoho Creator, applying the features covered in Units I–V, and documents it in this report.

Formatting Standards (Anna University Convention)

Element	Specification
Paper size / Margins	A4; Left 1.5" (binding side), Top 1", Right 1", Bottom 1"
Font	Times New Roman throughout; code in Courier New 10 pt
Body text	12 pt, justified, 1.5 line spacing
Chapter title	14 pt, bold, capitals, centred; every chapter starts on a new page
Section / Sub-section	12 pt bold (1.1, 1.1.1 numbering)
Figures	Centred; caption below — "Figure 4.2: ER Diagram of …"; numbered chapter-wise
Tables	Centred; caption above — "Table 5.1: Form Field Specification"; numbered chapter-wise
Page numbers	Front matter in roman (i, ii, iii…); Chapter 1 onwards in arabic (1, 2, 3…), bottom centre
References	IEEE style, numbered in order of citation [1], [2]…
Length	No page limit — the report grows with the application. Every feature you implement must be evidenced.

Working with the Template

- Text in [square brackets with yellow highlight] is a placeholder. Replace it, then remove the highlight.
- Sample rows in tables are illustrations. Replace them with your own application's data and add as many rows as needed.
- Grey dashed boxes are screenshot placeholders. Paste your screenshot inside the box (or replace the box) and keep the caption below it.
- Captions use the styles "FigCaption" and "TabCaption". Keep these styles when adding new figures or tables so they appear in the List of Figures / List of Tables.
- After editing, right-click the Table of Contents, List of Figures and List of Tables and choose Update Field → Update entire table (or press Ctrl+A, then F9).

- Sections that do not apply to your application (for example, 5.10 Pro-Code Extension) must still be retained with a short justification of why the feature was not used.
- Complete Appendix A (Feature Coverage Checklist) honestly; examiners verify it against the live application during the demonstration.

STUDENT SCHOLARSHIP & FINANCIAL AID MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

INDHUMATHY K

(241401030)

in partial fulfilment of the End Semester Examination for the course

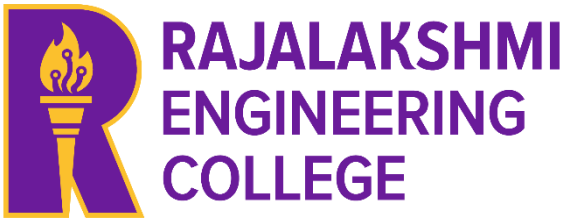
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND BUSINESS SYSTEM



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

OCT - 2026

RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this project report titled "STUDENT SCHOLARSHIP AND FINANCIAL AID MANAGEMENT SYSTEM" is the bonafide work of INDHUMATHY K (241401030), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year [2026–2027], Semester [V], under my supervision.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

I, **INDHUMATHY K (241401030)**, hereby declare that the project report titled **"STUDENT SCHOLARSHIP AND FINANCIAL AID MANAGEMENT SYSTEM",** submitted for the End Semester Examination of the course **CB23F35 – Low-Code Application Development using Zoho Creator**, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: [Chennai]
Date: [DD Month YYYY]

SIGNATURE OF THE STUDENT
INDHUMATHY K
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ACKNOWLEDGEMENT

I express my sincere gratitude to our Chairman **Mr. S. Meganathan, B.E., F.I.E.**, Chairperson **Dr. Thangam Meganathan, Ph.D.** and Vice-Chairman **Mr. Abhay Shankar Meganathan, B.E., M.S.**, for providing the infrastructure and facilities that made this work possible.

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INDHUMATHY K
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ABSTRACT

Problem context: Educational institutions manage scholarship and financial aid applications by collecting student details, checking eligibility, maintaining supporting documents, and tracking application progress. When these activities are handled manually or through separate spreadsheets, administrators may experience difficulties in organising records, retrieving information, monitoring application status, and completing repetitive tasks. These challenges can increase administrative workload and make the application process less organised. A centralised digital system can help simplify scholarship management and improve the accessibility of application information.

Solution: The Student Scholarship & Financial Aid Management System is proposed as a low-code application using Zoho Creator. The application will include forms for student information and scholarship applications, reports for viewing and managing records, and interfaces for monitoring application status. Deluge scripting and workflow automation will be configured where appropriate to support data validation, status updates, and notifications. Role-based access will be designed to separate student and administrator responsibilities and protect information according to user permissions. Integrations with Zoho services or external APIs will be considered based on the requirements and feasibility of the application.

Outcome: The system will be tested using functional and negative test cases to evaluate form validation, record management, application status tracking, workflow execution, and access permissions. Following successful testing, the application will be deployed using the available Zoho Creator deployment options. Performance, security settings, and access controls will be reviewed to ensure reliable operation. The final evaluation will measure improvements using actual test results, including reductions in manual steps and application processing time, where measurable. The project aims to demonstrate how low-code development can support efficient scholarship administration while providing opportunities for future enhancements.

Keywords: *Low-Code Development; Zoho Creator; Deluge Scripting; Workflow Automation; Role-Based Access Control; Scholarship Management*

PROJECT AT A GLANCE

Attribute	Details
Project Title	Student Scholarship and Financial Aid Management System
Application Domain	Education
Student Name & Register No.	Indhumathy K – 241401030
Zoho Creator Edition	Free
Application Name (Link Name)	Stud Scholarship & Financial Aid Mngt System
Published / Live URL	[https://creatorapp.zoho.in/…]
Demo Credentials (per role)	[Admin: … | Role 2: … | Role 3: …]
Number of Forms / Subforms	[ ] / [ ]
Number of Reports (by type)	[List:] [Kanban:] [Calendar:] [Pivot/Chart:] [Others:]
Number of Pages / Dashboards	[ ]
Number of Workflows	[Form:] [Schedule:] [Approval:] [Blueprint:] [Batch:]
Deluge Functions / Custom Actions	[ ]
Integrations	[e.g., Zoho CRM, Zoho Mail, REST API – OpenWeather]
Roles & Profiles Defined	[e.g., Administrator, Manager, Staff, Customer (Portal)]
Environments Used	[Development → Stage → Production]
Date of Production Deployment	[DD Month YYYY]

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LIST OF ABBREVIATIONS

> ***Guidance:*** *List every abbreviation used in the report in alphabetical order. Remove the ones you do not use.*

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
JSON	JavaScript Object Notation
LCAP	Low-Code Application Platform
OAuth	Open Authorization
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDLC	Software Development Life Cycle
SSO	Single Sign-On
UAT	User Acceptance Testing
UI	User Interface
Zia	Zoho’s AI assistant
[ ]	[ ]

CHAPTER 1
INTRODUCTION

1.1 Background

Scholarship and financial aid management is an important administrative activity in educational institutions. Scholarships help eligible students continue their education by providing financial assistance based on academic performance, family income, merit, or other eligibility criteria. Educational institutions may manage different scholarship schemes, each with its own application requirements, supporting documents, eligibility conditions, and approval procedures. Students need a convenient way to submit their information, while administrative staff require an organised method to review applications and maintain accurate records.

In a typical educational environment, the scholarship application process involves students submitting personal details, academic information, family income details, and supporting documents. Administrative staff collect these details, verify the submitted information, check eligibility requirements, and forward applications for review and approval. The application status must also be maintained so that students and authorised staff can identify whether an application is submitted, under review, approved, or rejected. When these activities are handled through paper forms, spreadsheets, or separate digital files, managing the information can become complicated, particularly when the number of applications increases.

The Student Scholarship and Financial Aid Management System is designed for an educational institution that requires a centralised method of managing student scholarship applications. The proposed system considers students and administrative personnel as its primary stakeholders. Students need to provide the required information accurately, while administrators need to organise records, review applications, monitor their progress, and maintain reliable information for future reference. A structured application can help reduce repetitive data entry, improve record accessibility, and support a more consistent application management process.

The project uses Zoho Creator, a low-code application development platform, to develop the proposed system. The application is planned to include student information forms, scholarship application forms, reports, and suitable interfaces for managing

application records. Relevant validation rules and workflow automation may also be incorporated to support the application process. The project focuses on demonstrating how low-code development can be applied to an educational administration problem through a practical, structured, and maintainable information management solution.

1.2 Problem Statement

Managing scholarship and financial aid applications using paper-based procedures or separate spreadsheets can make it difficult for educational institutions to maintain accurate student records, verify submitted information, monitor application progress, and organise approval activities. Students may experience uncertainty about their application status, while administrative staff may spend additional time locating records, checking details, and preparing reports. Therefore, a centralised Student Scholarship and Financial Aid Management System is proposed to organise application information, improve record accessibility, and support a more systematic scholarship management process.

- The major problems identified are:
- **Data duplication:** Student and scholarship details may be recorded in multiple files, increasing the possibility of inconsistent or duplicate information.
- **Time-consuming processing:** Manually collecting, verifying, and organising application details can increase administrative workload.
- **Limited status visibility:** Students and authorised staff may find it difficult to track the progress of individual scholarship applications.
- **Manual reporting:** Preparing summaries of submitted, approved, or rejected applications may require additional effort.
- **Access control challenges:** Without clearly defined permissions, controlling access to sensitive student and family financial information may be difficult.

1.3 Objectives

The primary objective of this project is to develop a Student Scholarship and Financial Aid Management System using Zoho Creator to organise student information and support scholarship application processing.

The specific objectives are:

- To design structured forms for collecting student details, academic information, and scholarship application data.
- To organise application records using suitable reports and data views for efficient retrieval and monitoring.
- To configure field validation and relevant Deluge scripting or workflows to reduce data-entry errors and automate selected tasks.
- To provide suitable interfaces for administrators to review applications and monitor their status.
- To configure appropriate user permissions to protect student information and restrict unauthorised access.
- To test the application's functional requirements and evaluate its effectiveness using actual test results and measurable observations.

1.4 Scope of the Project

The scope of the project covers the design and development of a low-code application for managing student scholarship and financial aid information using Zoho Creator. The system is intended to provide a centralised location for collecting student details, recording scholarship applications, organising application records, and monitoring application status.

In-scope activities:

- Creating forms for student information and scholarship application details.
- Defining suitable fields and validation rules for data collection.
- Organising application records through reports and relevant data views.
- Developing interfaces for viewing and managing application information.
- Configuring relevant workflows and Deluge scripts, where required and supported.
- Defining user permissions for administrative access and data protection.
- Testing form submissions, record retrieval, validation, status management, and implemented automation.
- Deploying the application using the available Zoho Creator options, subject to account and plan capabilities.

The primary intended users are students who provide application information and administrators who manage and review records. Separate student access, a customer

portal, or additional staff roles will be included only if they are implemented and supported by the available account configuration.

Out-of-scope activities:

- Direct transfer of scholarship funds or integration with banking and payment systems.
- Automatic verification of official income certificates through government databases.
- Integration with external services that are not configured and tested during the project.
- Development of a separate application using traditional programming technologies.
- Replacement of an institution's official scholarship policies, eligibility rules, or final approval authority.

The project is intended as an educational application prototype. Its actual functionality will depend on the features implemented and the capabilities available in the selected Zoho Creator edition.

1.5 Justification for a Low-Code Approach

A low-code development approach is suitable for the Student Scholarship and Financial Aid Management System because the application primarily requires structured data collection, record management, reports, and workflow automation. Developing these features entirely through traditional programming would require additional effort for designing interfaces, implementing data handling, creating business logic, and managing deployment. Zoho Creator provides visual development tools that can simplify these activities and reduce the amount of conventional coding required.

Zoho Creator supports the creation of forms, reports, pages, workflows, and business rules through its visual development environment. Deluge scripting can be used to implement additional logic when configuration alone is insufficient. These capabilities are relevant to the proposed application because student registration, scholarship application submission, record validation, and status management can be organised within a single platform. The platform also supports web and mobile access, integration capabilities, and configurable access permissions, subject to the selected plan and application configuration.

Another advantage of low-code development is maintainability. Changes to form fields, reports, and workflows can be made through the development interface without rebuilding the entire application from the beginning. This can be useful in an educational environment where scholarship requirements or administrative procedures may change. Zoho Creator also provides cloud hosting and deployment facilities, reducing the need to independently manage application infrastructure.

The choice of Zoho Creator is therefore justified by its visual development capabilities, support for workflow automation, integrated data management, and potential for rapid prototyping. However, the final implementation must be evaluated against the platform's plan limitations, security requirements, and actual application needs.

1.6 Organisation of the Report

This report is organised into nine chapters. Chapter 1 introduces the project, its problem statement, objectives, scope, and justification for using a low-code approach. Chapter 2 presents the literature survey and discusses related approaches and emerging trends in low-code development. Chapter 3 describes the requirements and analysis of the proposed system. Chapter 4 presents the system design, including architecture, data modelling, forms, and workflows.

Chapter 5 documents the implementation of the application using Zoho Creator, including forms, reports, pages, scripts, automation, integrations, and access control. Chapter 6 describes application testing, debugging, and the evaluation of functional requirements. Chapter 7 discusses deployment, performance optimisation, security, and governance. Chapter 8 presents the results, limitations, and discussion of the developed system. Chapter 9 concludes the report and identifies possible future enhancements. References and appendices provide supporting sources, code, checklists, and user documentation.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In a conventional scholarship and financial-aid management process, students obtain information about available scholarships through college notices, institutional websites, department announcements, or financial-aid offices. They identify suitable schemes and submit applications with personal, academic, family-income, and other supporting information. The responsible staff members then collect the applications and verify the submitted details and documents.

After verification, eligible applications are forwarded to the appropriate authority for review and approval. The application status and award details are communicated to students through the communication methods used by the institution. Staff members may also maintain records of applications received, documents verified, applications approved or rejected, and the amount of financial aid awarded.

When information is maintained in separate forms, spreadsheets, emails, or physical files, staff may need to consolidate data manually to prepare reports. Students may also need to contact the concerned office to learn about the status of their applications.

The exact process may vary between institutions. The following diagram represents a typical existing process considered for this project.

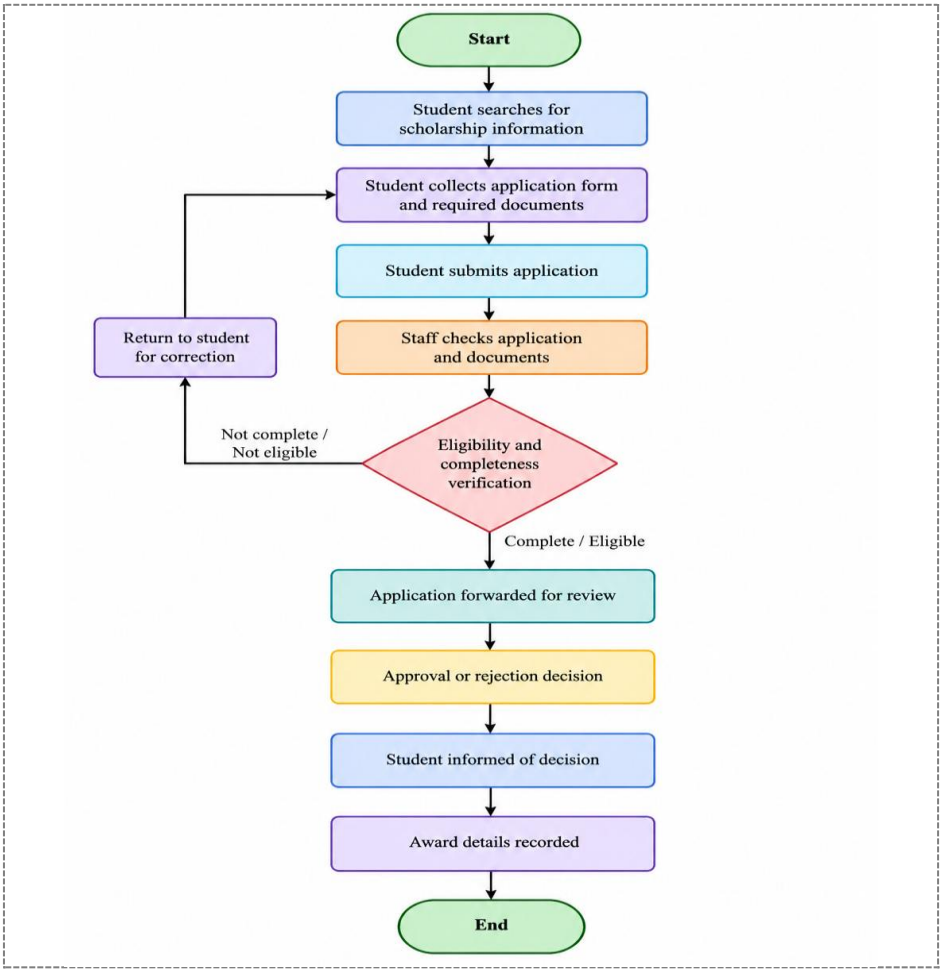


Figure 2.1: Process flow of the existing system

2.2 Limitations of the Existing System

- **Scattered information:** Student details, scholarship information, supporting documents, and application status may be stored in different files or systems, making it difficult to retrieve complete information quickly.

- **Time-consuming verification:** Manual checking of application details and documents can increase the time required to process applications and may lead to repeated work.

- **Limited status visibility:** Students may have to contact the scholarship office to find out whether an application has been received, is under verification, or has been approved or rejected.

- **Data duplication and errors:** Re-entering information into multiple records can result in inconsistent details, duplicate entries, or accidental mistakes.

- **Difficult reporting:** Preparing summaries of applications, eligibility outcomes, approved awards, and financial-aid amounts may require manual consolidation.

- **Access and tracking challenges:** Without clearly defined permissions and reliable record tracking, it can be difficult to ensure that users access only the information relevant to their responsibilities.

These limitations describe the problems that the proposed system aims to address. Their actual occurrence and severity should be confirmed against the process followed by the institution.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Requires implementation of the user interface, database, business logic, authentication, and reports. A preliminary planning estimate might be 6–10 weeks for a small academic prototype, depending on scope and experience.	Provides visual form, report, and workflow builders. A preliminary planning estimate might be 2–4 weeks for a comparable prototype, depending on scope and familiarity with the platform.
Skills required	Requires programming, database design, testing, and deployment knowledge.	Primarily requires application design and configuration skills, with Deluge scripting useful for custom logic.
Hosting & infrastructure	The developer must select and configure hosting, database services, backups, and related infrastructure, or use managed services.	The application runs on Zoho Creator's managed platform, subject to the selected plan and platform capabilities.
Mobile application	Mobile support must be designed and implemented or provided through a suitable framework.	Zoho Creator supports mobile access to applications; the exact user experience depends on configuration and platform capabilities.
Security & access control	Authentication, authorisation, validation, and audit-related features must be designed and implemented or integrated.	Platform permissions and sharing settings can be configured; they must still be designed, tested, and reviewed for the application.

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Maintenance & change	Changes may require modifying code, tests, database structures, and deployment configurations.	Many changes can be made through the visual builder; complex changes may still require scripts or redesign.
Customisation limits	Offers broad control over the technology stack and implementation, subject to time, skills, and budget.	Customisation depends on Creator's features, supported integrations, Deluge, plan limits, and platform constraints.
Cost	May involve developer time, hosting, database services, maintenance, and other tools.	May reduce initial development effort, but subscription charges, plan limits, integrations, and ongoing costs must be considered.

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Bock, A. C., and Frank, U. (2021)	*Low-Code Platform*	Discusses the characteristics of low-code platforms, their relationship to established software-development approaches, and their potential opportunities and limitations.	Provides a research basis for selecting and evaluating a low-code approach rather than assuming it is suitable for every requirement.

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
2	Waszkowski, R. (2019)	*Low-code platform for automating business processes in manufacturing*	Describes the use of a low-code platform to automate business processes and discusses low-code as an approach to delivering business applications with reduced conventional coding effort.	Supports investigating whether application forms and administrative processes can be modelled and automated using a low-code platform.
3	Zoho Corporation (n.d.)	*Understand Application — Zoho Creator Help*	Explains how forms collect and store information, reports present records, and workflows perform actions in response to events.	Helps guide the design of student, scholarship, and financial-aid records, their reports, and their automated actions.
4	Zoho Corporation (n.d.)	*Creating and Managing Form Workflows — Zoho Creator Help*	Explains how workflows can be triggered by form and record events and configured to perform actions such as updating information, sending notifications, and calling integrations.	Supports the design of application-submission checks, status updates, and notifications, where configured and tested.

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
5	Zoho Corporation (n.d.)	*Creating Applications with AI Using Zia App Builder — Zoho Creator Help*	Describes AI-assisted application creation from natural-language requirements and explains that generated components must be reviewed and refined. Availability depends on the applicable Creator plan and configuration.	Provides background for evaluating AI-assisted low-code development. It does not mean AI features have been used in this project.

The research paper by Bock and Frank provides a conceptual discussion of low-code platforms, while Waszkowski presents a practical application of low-code process automation in a different domain. Zoho Creator documentation provides implementation-level information about forms, reports, workflows, and AI-assisted application creation. Together, these sources support examining the suitability of low-code tools for an administrative information system.

However, the scholarship-management requirements of this project must be established independently. The selected platform and literature do not by themselves prove that eligibility decisions, financial calculations, security, or notifications will work correctly. Those functions must be specified, implemented where possible, and tested.

2.5 Emerging Trends in Low-Code Development

AI-assisted application building: Low-code platforms increasingly provide features that use natural-language descriptions to help generate application components or scripts. In a scholarship-management application, such tools could help create an initial form structure or suggest a workflow. Any generated components would still need to be checked against the requirements. AI-assisted features are not assumed to have been used in this project unless they are actually tested.

Citizen development: Visual builders allow users with limited conventional programming experience to configure forms, reports, and basic workflows. This can help administrative teams participate in designing a system that reflects their process. Appropriate reviews and access controls remain important, especially when handling student information.

Hyper-automation: This trend combines multiple automated steps, such as validation, notifications, approval routing, and record updates. For the proposed system, simple event-based workflows could reduce repetitive administrative tasks. More advanced automation should be added only where it is supported by the platform and the process requirements.

Composable applications and integration: Applications can be assembled from separate forms, reports, workflows, reusable functions, and external services. For this project, a modular design can separate student records, scholarship details, applications, verification, and award tracking. Integrations with email or other services may extend the system, but must be configured and tested before being reported as implemented.

These trends influenced the proposed design by favouring modular data collection, reusable workflow logic, clear reports, and the possibility of extending the application later. The practical implementation will determine which of these features are demonstrated in the final system.

2.6 Proposed System

The proposed Student Scholarship & Financial Aid Management System will be developed using Zoho Creator as a centralised application for organising scholarship and financial-aid information. The planned system will provide a structured way to record student details, maintain scholarship information, submit applications, review application records, and track decisions and award details.

The planned application may include the following components:

1. **Student Information:** A form to record essential student details and information needed for scholarship processing.

2. **Scholarship Details:** A form to maintain scholarship names, eligibility conditions, application periods, and other relevant scheme information.

3. **Scholarship Application:** A form that associates a student with a scholarship and records application details and status.

4. **Verification and Review:** Records or workflow steps to support completeness checks, eligibility review, and approval or rejection decisions.

5. **Reports and Dashboard:** Views to help authorised users track submitted applications, pending reviews, decisions, and award totals.

6. **Workflow Automation:** Where configured, rules may validate entries, update statuses, and send relevant notifications.

7. **Access Control:** Permissions will be designed to limit access according to user responsibilities and will be tested before claiming that the system is secure.

How the proposed system addresses the limitations

Existing-system limitation	Proposed response
Scattered information	Store related student, scholarship, and application records in one organised application.
Time-consuming verification	Use structured fields and, where suitable, validation rules to reduce avoidable manual checking.
Limited status visibility	Maintain an application-status field and provide appropriate reports for authorised users.
Data duplication and errors	Use consistent fields and lookup relationships where appropriate; validate required data.
Difficult reporting	Create filtered reports and summary views based on stored application records.
Access and tracking challenges	Configure role-based permissions and test whether each role can access only the intended information.

The system is intended to improve the organisation and traceability of scholarship-related information. The actual benefits will depend on the components implemented and the results of testing. Claims about time savings, error reduction, external integration, security, or deployment should be made only when supported by evidence from the completed application.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	User responsible for maintaining the application and its configuration.	Manage users and permissions, configure settings, maintain scholarship records, and access authorised data.
Manager	User responsible for reviewing scholarship applications and making decisions.	Review verified applications, approve or reject applications, and view application summaries and dashboards.
Staff / Operator	User responsible for day-to-day data entry and application processing.	Register student details, maintain scholarship information, check application records, and update verification status.
Student / Portal User	Student who applies for financial assistance.	Provide required student details, submit scholarship application information, and view application status if student-facing access is configured.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow authorised staff to register a student by entering the mandatory student details, including student name, register number, email address, department, programme, and year of study.	Administrator, Staff / Operator	[High]
FR-02	The system shall allow authorised users to create and maintain scholarship records containing the scholarship name, eligibility criteria, application deadline, and available award details.	Administrator, Staff / Operator	[High]
FR-03	The system shall allow staff to create a scholarship application linked to an existing student record and scholarship record.	Staff / Operator	[High]
FR-04	The system shall allow authorised staff to record document-verification status and eligibility-review status for each application.	Staff / Operator	[High]
FR-05	The system shall allow the manager to record an approval or rejection decision and maintain the application status.	Manager	[High]

FR ID	Requirement	Role(s)	Priority
FR-06	The system shall allow authorised users to record the financial-aid amount and award details for an approved application.	Manager, Staff / Operator	[High]
FR-07	The system shall display application records filtered by status, such as Submitted, Under Review, Approved, and Rejected.	Administrator, Manager, Staff / Operator	[High]
FR-08	The system shall provide a report or dashboard summarising application counts by status.	Administrator, Manager	[Medium]
FR-09	The system shall validate mandatory fields and prevent submission when required information is missing, where supported by the configured form.	Administrator, Staff / Operator	[High]
FR-10	The system shall send an email notification when an application decision is recorded, if email integration and the required workflow are configured.	Manager; Student as recipient	[Medium]
FR-11	The system shall restrict access to application functions and records according to configured user permissions.	Administrator, Manager, Staff / Operator, Student / Portal User	[High]
FR-12	The system may synchronise selected application information with Zoho CRM or another external service if an integration is configured and tested.	Administrator	[Medium]

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	For the academic prototype, reports containing up to 500 sample records should load within 5 seconds under normal test conditions. This target must be measured during testing.
NFR-02	Security	Users shall access only the application functions and records permitted by their assigned role and configured permissions.
NFR-03	Usability	The application forms and reports should be usable through a supported web browser and, if supported and tested, the Zoho Creator mobile application.
NFR-04	Availability	The application shall be accessible when the Zoho Creator service is available and the user has a working internet connection and valid access.
NFR-05	Maintainability	Forms, reports, and workflows shall use clear names and a consistent structure so that authorised changes can be made and tested.
NFR-06	Auditability	Where platform audit or history features are available and enabled, relevant record changes should be traceable through the available audit information.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator; the available edition and features depend on the account plan.
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment), where scripting is required.
Browser	A current supported version of Google Chrome, Microsoft Edge, or Mozilla Firefox.
API testing tool	Postman
Other Zoho services	Zoho Mail, Zoho CRM, Zoho Analytics, or Zoho Catalyst only if required and configured for the project.
Mobile	Zoho Creator mobile application for Android or iOS, if mobile testing is included.
Client hardware	Computer with an Intel Core i3/i5 or equivalent processor, 8 GB RAM, and a reliable internet connection. These are suggested project-use specifications, not mandatory Zoho Creator platform requirements.

3.5 Use Case Model

The use case model describes how the different users interact with the Student Scholarship & Financial Aid Management System.

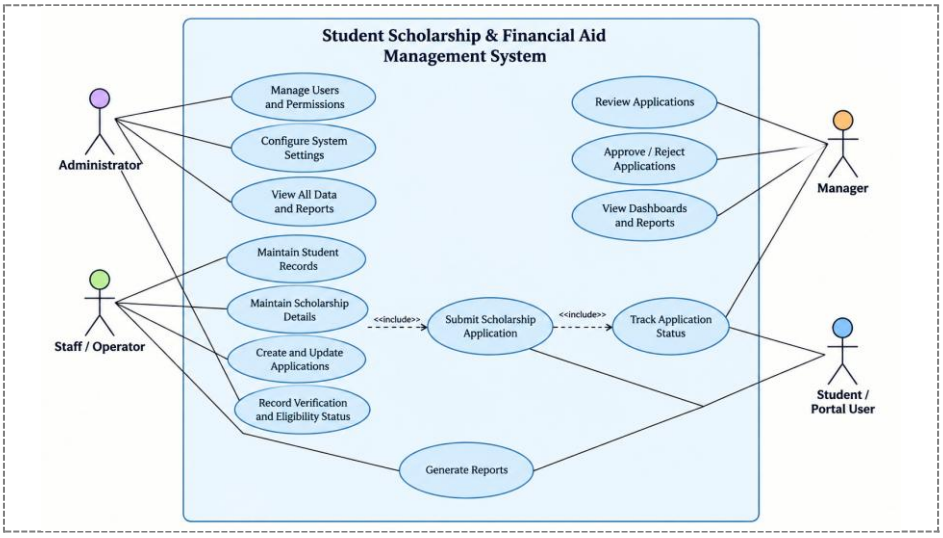


Figure 3.1: Use Case Diagram of the Student Scholarship and Financial AID Management System

Table 3.5: Use Case Description

Use Case ID	UC-01
Use Case Name	Register Student
Actor(s)	Administrator, Staff / Operator
Pre-condition	The user is authorised to access the student form.
Main Flow	1. The user selects the student and scholarship. 2. The user enters the required application information. 3. The user submits the application. 4. The system checks mandatory fields. 5. The system saves the application and assigns its initial status.
Alternate Flow	If mandatory information is missing, the system requests correction. If student-facing access has not been configured, an authorised staff member enters the application instead.
Post-condition	The application is saved and can be tracked by authorised users.
Related FR	FR-03, FR-07, FR-09, FR-11

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

The Student Scholarship & Financial Aid Management System follows a layered architecture. It uses Zoho Creator as the low-code application platform to manage student details, scholarship information, scholarship applications, and application reports.

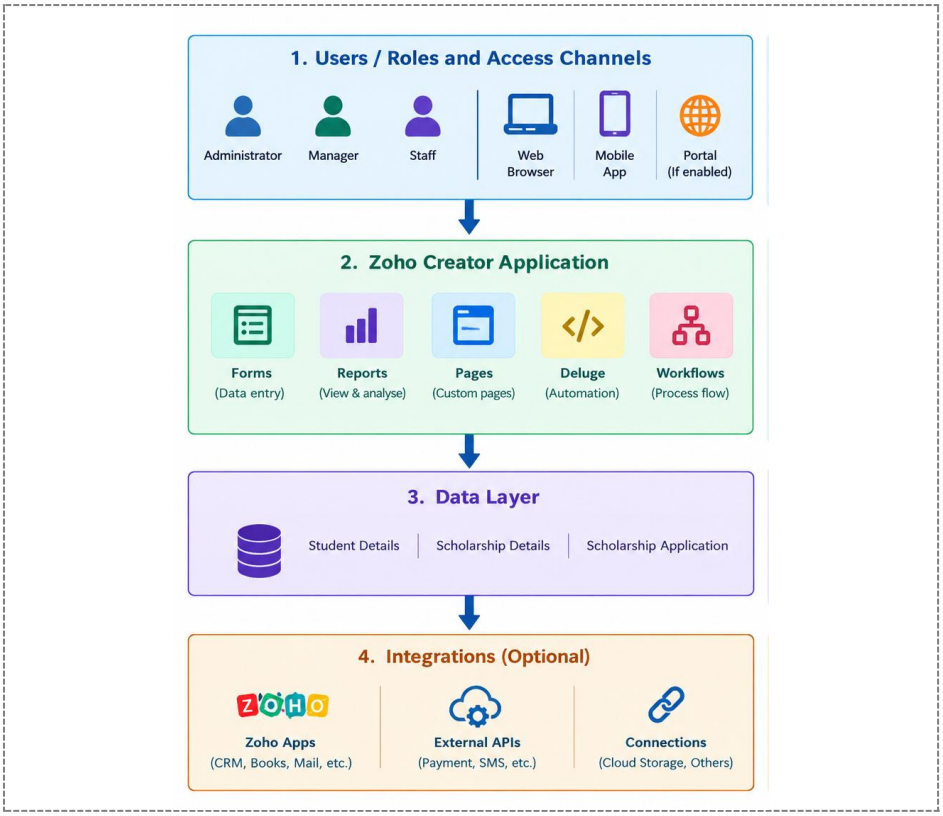


Figure 4.1: System Architecture of the Student Scholarship & Financial Aid Management System

The architecture consists of the following layers:

1. User Layer: Administrators, managers, and staff access the application through a web browser.

2. Application Layer: Zoho Creator provides forms, reports, and application interfaces to manage student records, scholarship details, and applications.

3. Business Logic Layer: Application rules and status fields support the management and review of scholarship applications. Deluge scripts and automated workflows can be added when required.

4. Data Layer: The application stores student details, scholarship details, and scholarship application records within Zoho Creator.

5. Integration Layer: The application can be extended to connect with other Zoho services or external APIs when required.

4.2 Data Model

The data model defines the entities used in the application and the relationships between them. The system contains three main entities: Student Details, Scholarship Details, and Scholarship Application.

The Student Details entity stores information about students. The Scholarship Details entity stores information about available scholarships. The Scholarship Application entity connects a student with a scholarship and stores application-related information.

Lookup fields are used to establish relationships between forms in Zoho Creator. The Student Details and Scholarship Details lookup fields in the Scholarship Application form allow an application to reference the corresponding student and scholarship.

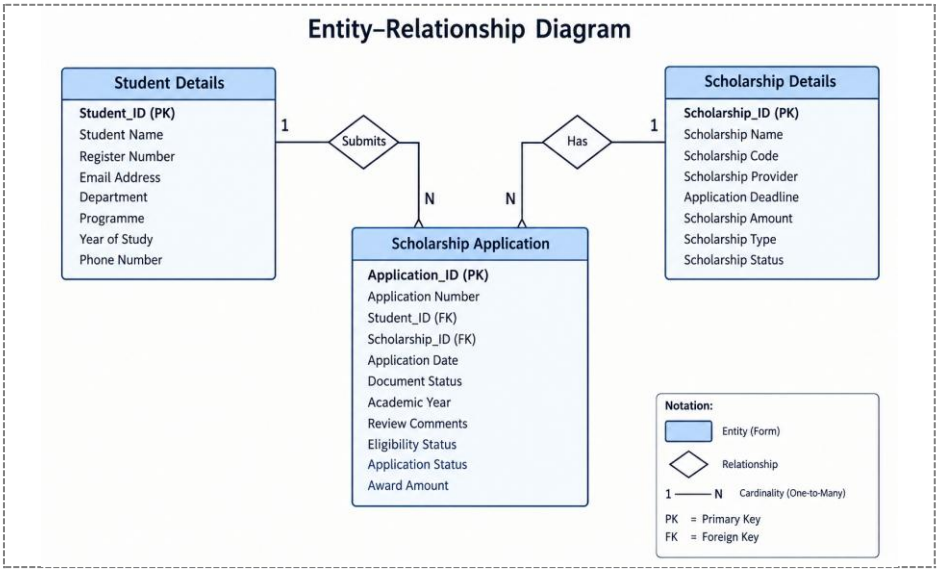


Figure 4.2: Entity–Relationship Diagram

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Student Details	Student Name, Register Number, Email Address, Department, Programme, Year of Study, Phone Number	Scholarship Application	One-to-Many	Lookup in Scholarship Application
Scholarship Details	Scholarship Name, Scholarship ID, Scholarship Provider, Application Deadline, Scholarship Amount, Scholarship Type, Scholarship Status	Scholarship Application	One-to-Many	Lookup in Scholarship Application
Scholarship Application	Application Number, Student Details, Scholarship, Application Date, Document Status, Academic Year, Review Comments, Application Status	Student Details and Scholarship Details	Many-to-One to each parent form	Lookup fields

4.3 Form Design

The application contains three main forms. Each form is designed to capture the information required for scholarship administration. The following tables describe the

field specifications. Link names and mandatory settings should be checked in Zoho Creator before final submission.

Table 4.2: Field Specification – Student Details Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Student Name	Verify in Creator	Single Line	Verify	Text entry	Student's name
Register Number	Verify in Creator	Single Line	Verify	Text entry	Unique student identifier
Email Address	Verify in Creator	Email	Verify	Email format	Student's email
Department	Verify in Creator	Single Line	Verify	Text entry	Department name
Programme	Verify in Creator	Single Line	Verify	Text entry	Academic programme
Year of Study	Verify in Creator	Drop Down	Verify	Select an available year	Current year of study
Phone Number	Verify in Creator	Phone	Verify	Phone format	Contact number

Table 4.3: Field Specification – Scholarship Details Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Scholarship Name	Verify in Creator	Single Line	Verify	Text entry	Name of scholarship

Scholarship Code	Verify in Creator	Single Line	Verify	Text entry	Scholarship identifier
Scholarship Provider	Verify in Creator	Single Line	Verify	Text entry	Providing organization
Description	Verify in Creator	Multi Line	Verify	Text entry	Scholarship description
Eligibility Criteria	Verify in Creator	Multi Line	Verify	Text entry	Eligibility requirements
Application Deadline	Verify in Creator	Date	Verify	Select a date	Last date to apply
Scholarship Amount	Verify in Creator	Currency or Number	Verify	Numeric amount	Scholarship value
Scholarship Type	Verify in Creator	Drop Down	Verify	Select a type	Type of scholarship
Scholarship Status	Verify in Creator	Drop Down	Verify	Select a status	Current availability

Table 4.4: Field Specification – Scholarship Application Form

Field Label	**Link Name**	**Field Type**	**Mandatory**	**Validation / Default**	**Remarks**
Application Number	Verify in Creator	Auto Number or configured field type	Verify	As configured	Application identifier
Student Details	Verify in Creator	Lookup	Verify	Select a student record	Links to Student Details

Scholarship	Verify in Creator	Lookup	Verify	Select a scholarship record	Links to Scholarship Details
Application Date	Verify in Creator	Date	Verify	As configured	Date of application
Document Status	Verify in Creator	Drop Down	Verify	As configured	Tracks document verification
Academic Year	Verify in Creator	Single Line	Verify	Enter academic year	Academic period
Review Comments	Verify in Creator	Multi Line	Verify	Optional comments if configured	Review remarks
Eligibility Status	Verify in Creator	Verify in Creator	Verify	As configured	Eligibility result
Application Status	Verify in Creator	Drop Down	Verify	Submitted, if configured as default	Submitted, Under Review, Approved, Rejected
Award Amount	Verify in Creator	Currency or Number	Verify	As configured	Awarded amount, if applicable

4.4 Process and Workflow Design

The scholarship application process begins when a student record and scholarship record are selected in the Scholarship Application form. The application information is entered and stored for review. The application status indicates the current stage of processing.

The basic process consists of the following steps:

1. Enter or select the student details.
2. Select the scholarship for which the student is applying.
3. Enter the application date, academic year, and document information.
4. Save the scholarship application.
5. Review the application and update its status when appropriate.
6. Record review comments and the award amount when applicable.

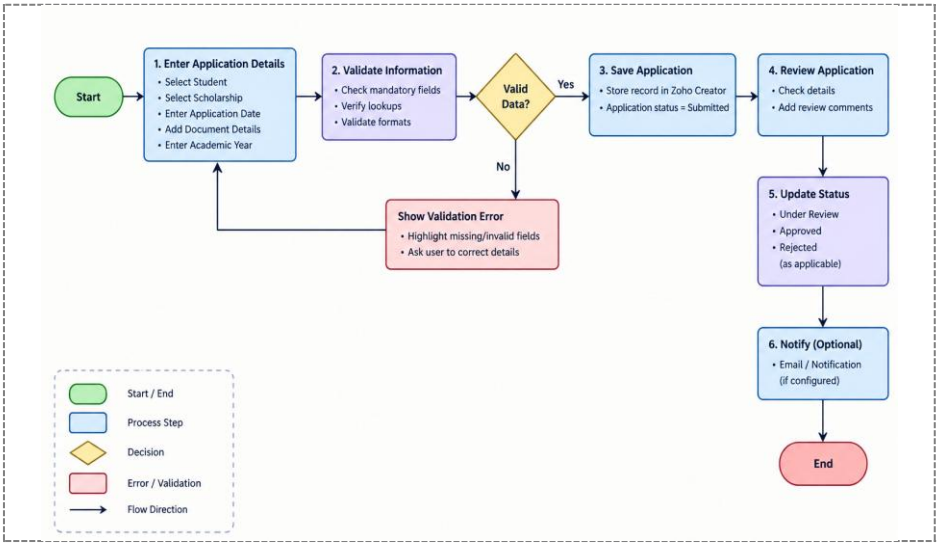


Figure 4.3: Workflow / Process Flow for Scholarship Application

4.5 User Interface and Navigation Design

The application uses a simple navigation structure to provide access to student information, scholarship records, and scholarship applications. The forms are designed to support data entry, while the reports allow users to view saved application records.

The main navigation components are:

- Student Details: Used to add and maintain student information.
- Scholarship Details: Used to maintain scholarship information and eligibility criteria.
- Scholarship Application: Used to create and update scholarship applications.
- Scholarship Application Details: Used to view saved application records.

- All Scholarship Application: Pivot Table report created to summarize application information.

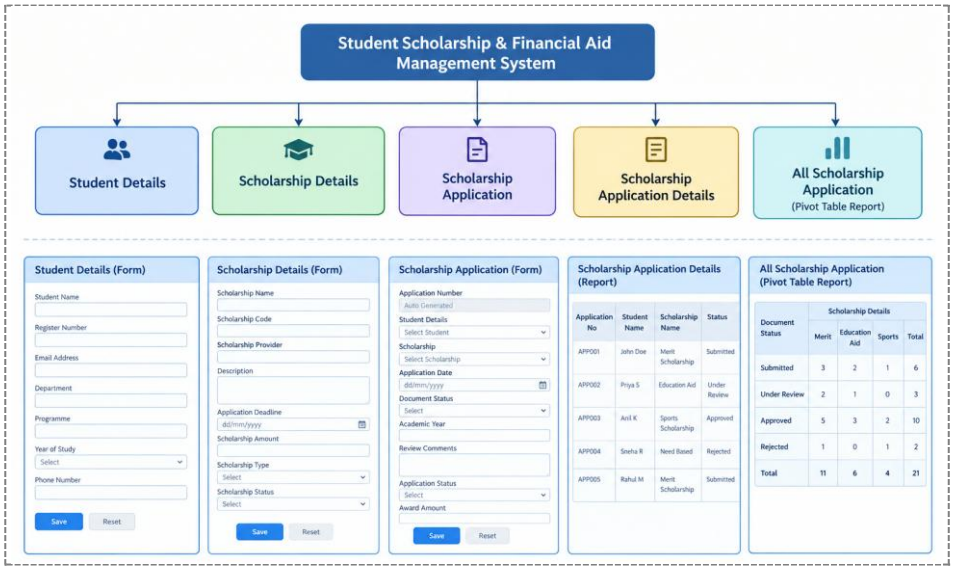


Figure 4.4: Navigation Structure and Wireframe of the application

4.6 Access Control Design

Access control determines which users can view, add, edit, or delete application information. The proposed role-based permission structure separates administrative tasks from application review and data entry.

Table 4.5: Role–Permission Matrix

Component	Administrator	Manager	Staff	Portal User
Student Details Form	A E V D	V	A E V	—
Scholarship Details Form	A E V D	A E V	V	—
Scholarship Application Form	A E V D	A E V	A E V	A, V (own records, if portal access is configured)
Scholarship Application Details Report	V	V	V	V (own records, if configured)
All Scholarship Application Pivot Table	V	V	V	—

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup.

The Student Scholarship and Financial Aid Management System was developed using Zoho Creator, a low-code application development platform. The application was created to simplify the management of student information, scholarship details, and scholarship applications.

The application contains three main forms: Student Details, Scholarship Details, and Scholarship Application. Reports were created to display and review the stored application records. The application is accessed through the Zoho Creator web interface.

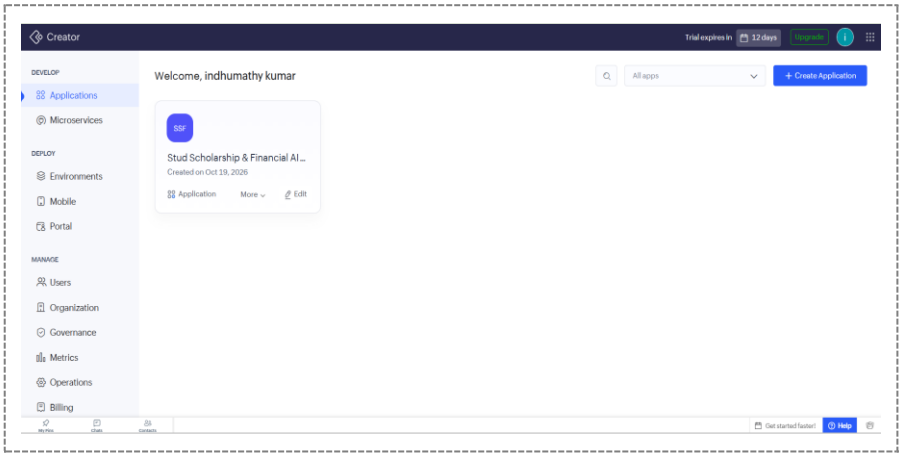


Figure 5.1: Application Home and Navigation Sections

5.2 Forms and Data Capture

Three forms were created to store and manage the information required for scholarship administration.

5.2.1 Student Details Form

The Student Details form stores the basic information of students, including their name, register number, email address, department, programme, year of study, and phone number. This information can be referenced when creating a scholarship application.

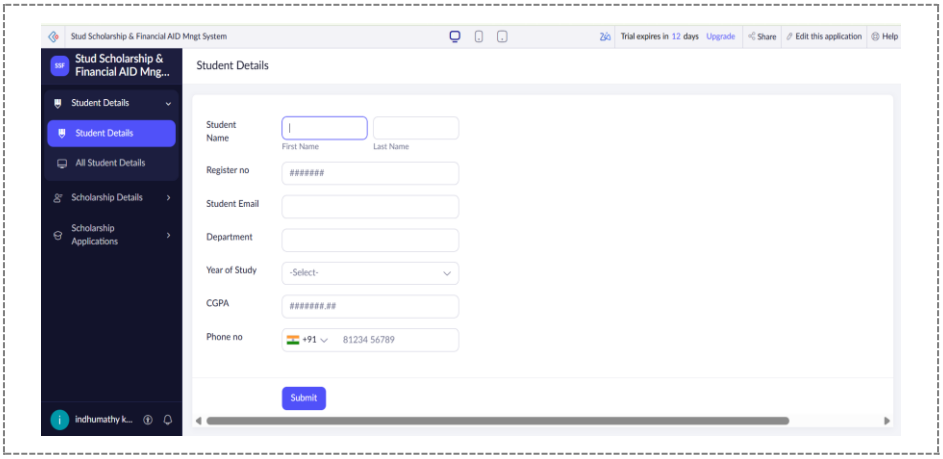


Figure 5.2.1a: Student Details Form in Live Mode

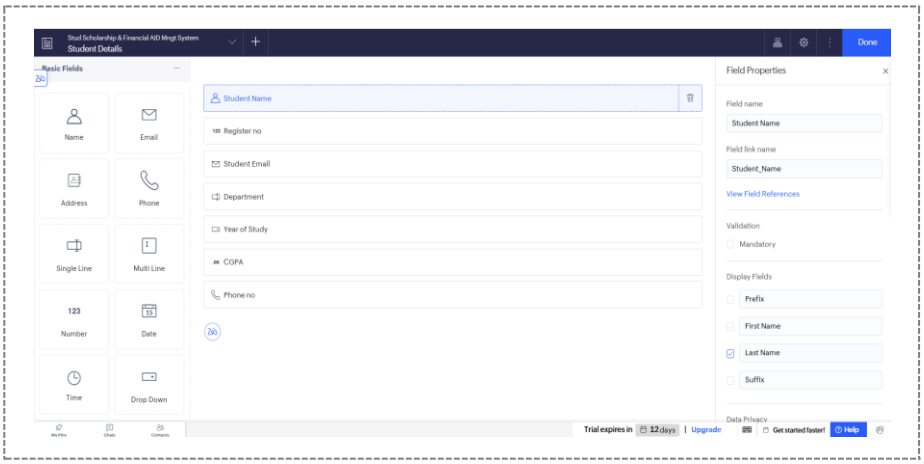


Figure 5.2.1b: Student Details Form in the Form Builder

5.2.2 Scholarship Details Form

The Scholarship Details form maintains information about available scholarships. It includes the scholarship name, scholarship code, provider, description, eligibility criteria, application deadline, scholarship amount, scholarship type, and scholarship status.

This form allows scholarship information to be stored in a structured manner for use during the application process.

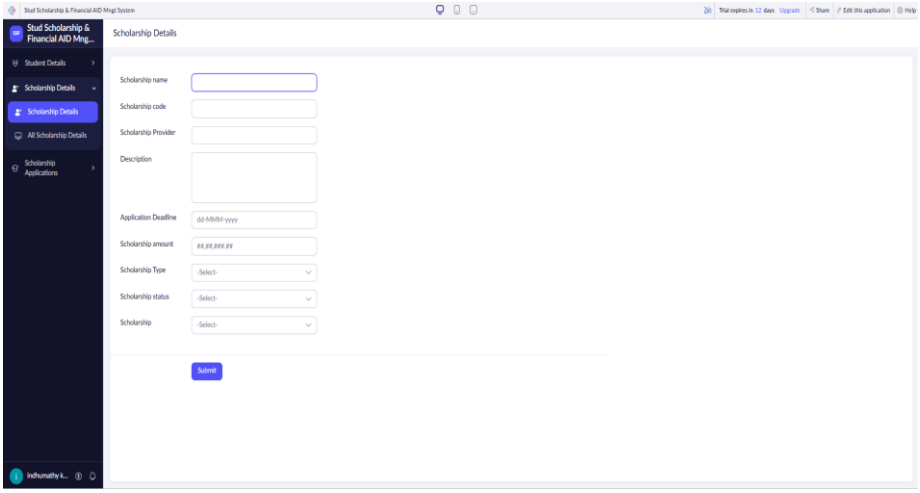


Figure 5.2.2a: Scholarship Details Form in Live Mode

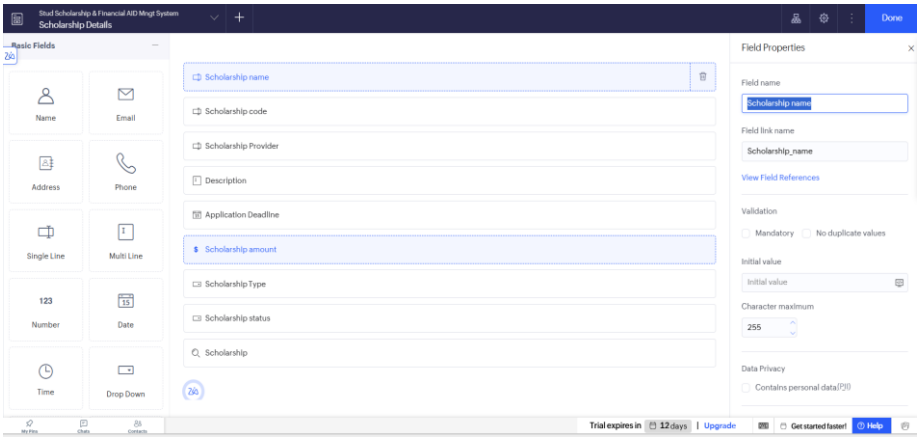


Figure 5.2.2b: Scholarship Details Form in the Form Builder

5.2.3 Scholarship Application Form

The Scholarship Application form connects student information with scholarship information through lookup fields. It stores the application number, application date, document status, academic year, review comments, eligibility status, application status, and award amount, according to the configured fields.

The Application Status dropdown includes Submitted, Under Review, Approved, and Rejected. This helps represent the progress of an application.

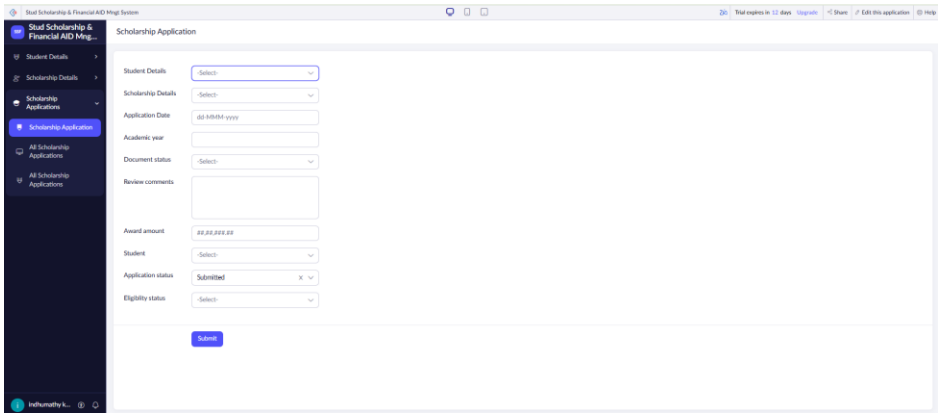


Figure 5.2.3a: Scholarship Application Form in Live Mode

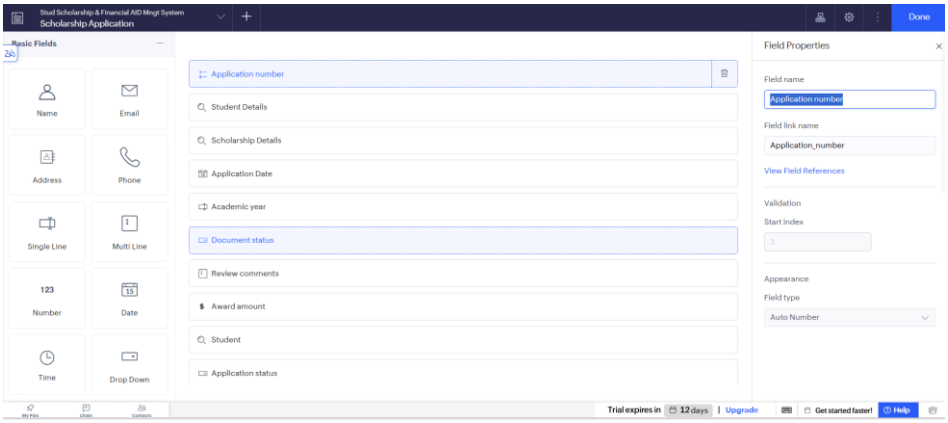


Figure 5.2.2b: Scholarship Application Form in the Form Builder

5.3 Reports

Reports are used to view and review records stored in the application. A list report displays scholarship application records, while a Pivot Table report was created to summarize application information.

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
Scholarship Application Details	List / Table	Scholarship Application	Displays saved application records	Default report actions	Authorized application users
All Scholarship Application	Pivot Table	Scholarship Application	Configured using document status and scholarship details	Default report actions	Authorized application users

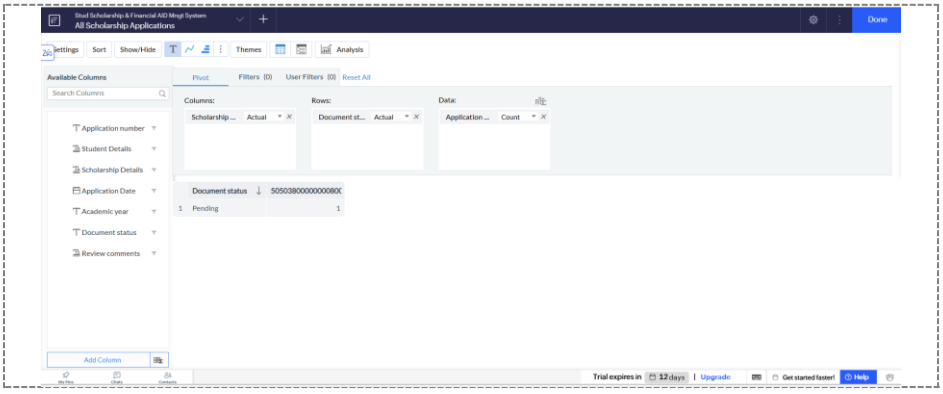


Figure 5.4: Scholarship Application Report

5.4 Pages and Dashboards

A custom page named **Scholarship Dashboard** was created using the Zoho Creator Page Builder. The page provides a visual overview of the Student Scholarship and Financial Aid Management System.

The dashboard contains a heading panel, summary cards, charts, an application status overview, and scholarship-related visual components. These elements are arranged to present scholarship application information in a single interface.

The dashboard is designed to help users understand application status and view scholarship-related information. The displayed counts and chart values should be verified against the saved application records.

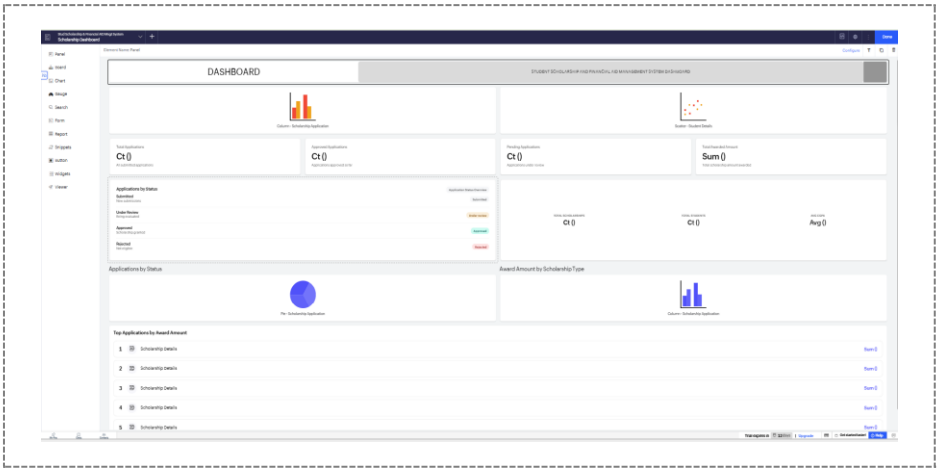


Figure 5.5: Student Scholarship and Financial AID Management System Dashboard

5.5 Deluge Scripting

A Deluge validation script was configured in the Scholarship Application form workflow. The script checks whether important application details, including eligibility criteria, student details, scholarship details, and application date, are provided. It also checks whether an award amount is entered when the application status is Approved.

The script was saved in Zoho Creator. Its validation behaviour must be tested using valid and incomplete application records before confirming the result.

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
[DS-01]	Scholarship Application – On Validate	Validate required fields and award amount for approved applications	`if`, `isNull()`, field validation
[DS-02]	Scholarship Application – On Success	Send an email notification after a new application is submitted	`Email notification`, `workflow action`

Code 5.1: Validation script

```
// Validate that the Application_Status field is not null or empty
if(input.Application_number1.isNull())
{
alert(Application_number1,"Please select an application status");
}
// Validate that the Eligibility_Criteria field is not null or empty
if(input.Eligiblity_Criteri.isNull())
{
alert(Eligiblity_Criteri,"Please select the eligibility criteria");
}
// Validate that the Award_Amount is not null or empty if status is
Approved
if(input.Application_number1 == "Approved")
{
if(input.Award_amount.isNull())
{
alert(Award_amount,"Please enter the award amount for approved
applications");
}
}
// Validate that Student_Details lookup is not null
if(input.Student_Details.isNull())
{
alert(Student_Details,"Please select a student");
}
// Validate that Scholarship_Details lookup is not null
if(input.Scholarship_Details.isNull())
{
alert(Scholarship_Details,"Please select a scholarship");
}
// Validate that Application_Date is not null
if(input.Application_Date.isNull())
{
alert(Application_Date,"Please enter the application date");
}
```

Code 5.2: Notification email

- Trigger: Scholarship Application record created successfully.
- Action: Send email.
- Recipient: Admin's email address.
- Subject: New Scholarship Application Submitted.
- Message: A new scholarship application has been submitted. Please review the application in Zoho Creator.

The validation workflow was configured to check required application fields. A separate notification workflow was configured to send an email when a scholarship

application was submitted successfully. The notification was tested, and the email was received by the configured recipient.

5.6 Workflows and Automation

Workflows were configured in Zoho Creator to support scholarship application validation and notification. The validation workflow uses a Deluge script to validate form submissions, while the notification workflow sends an email following successful form submission.

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Validate Scholarship Application	Form Workflow	Form validation	Configured validation rules	Validate submission and prevent invalid entries	DS-01
Notify Scholarship Application	Form Workflow	Successful form submission	Successful submission	Send email notification	—

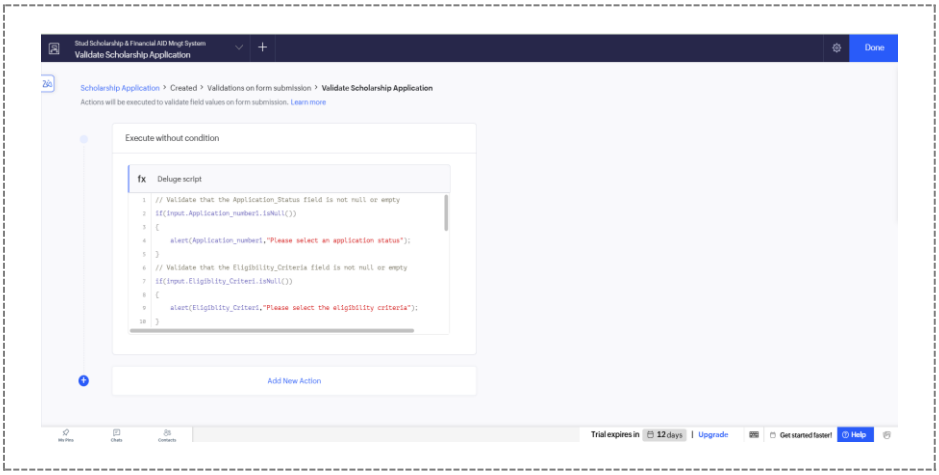


Figure 5.6.1a : Validate Scholarship Application

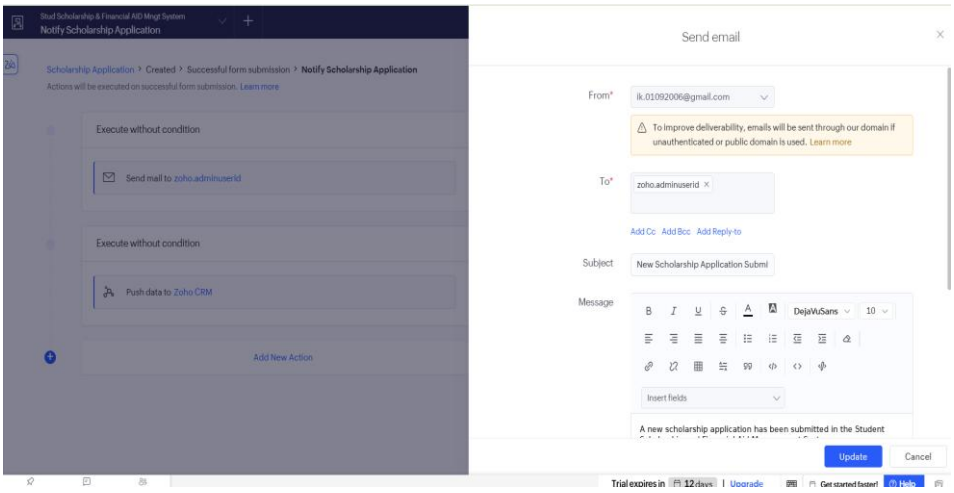


Figure 5.6.1b : Notify Scholarship Application

5.7 Integrations

The application integrates Zoho Creator with Zoho CRM to store scholarship application information in the CRM system. A connection named `scholarship_crm_connection` was created and enabled in Zoho Creator. The connection was selected in the workflow integration action, and the Create Lead action was configured to transfer the submitted application data.

The integration was tested by submitting a test scholarship application. The corresponding record appeared successfully in Zoho CRM, confirming that the configured integration was operational for the tested submission.

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Zoho CRM Connection	Zoho CRM	Creator Connection	Zoho authorization	Connection created and enabled; record-transfer testing remains to be verified

Code 5.3: Zoho CRM Integration Configuration

```
The Zoho CRM integration was configured using the Zoho Creator
integration action builder. The existing connection
scholarship_crm_connection was selected, and a Create Lead action was
configured to transfer scholarship application data to Zoho CRM.

The integration was tested by submitting a test scholarship
application. The corresponding record was successfully created in Zoho
CRM.
```

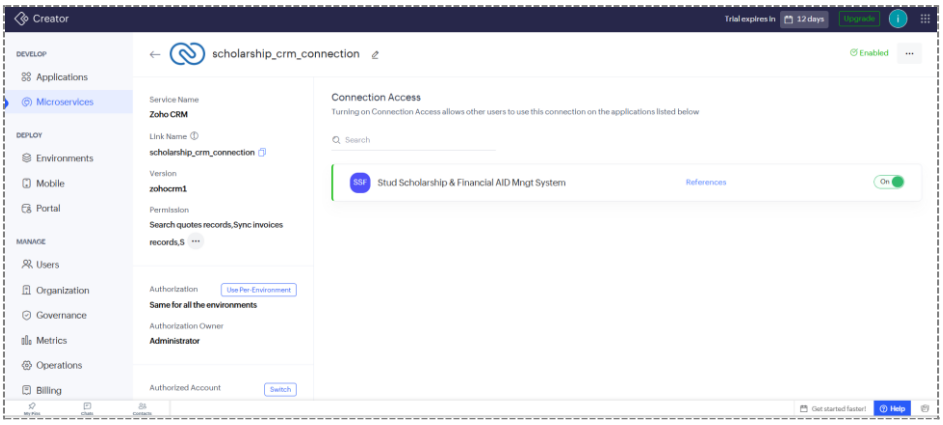


Figure 5.7: Zoho CRM Connection Configuration.

5.8 User Management and Role-Based Access

User access is managed through Zoho Creator's permission profiles, role hierarchy, and data-sharing settings. The application includes Read and Write permission profiles. The role hierarchy includes the CEO role, and a Scholarship Reviewer role was added to support the intended access structure.

A data-sharing rule was configured for scholarship application records to define access between roles. The selected access level and rule behaviour should be verified in the application before describing the rule as tested.

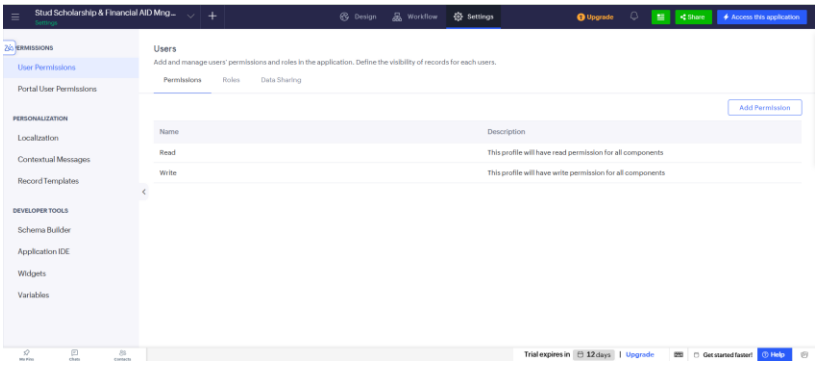


Figure 5.8.1a: Permission Profiles

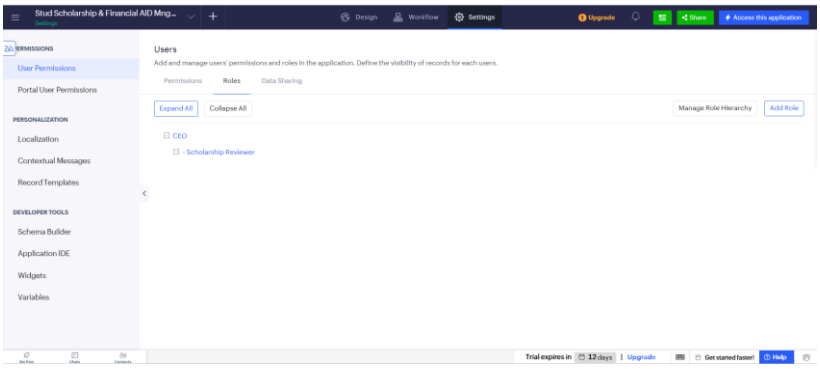


Figure 5.8.1b: Role Hierarchy

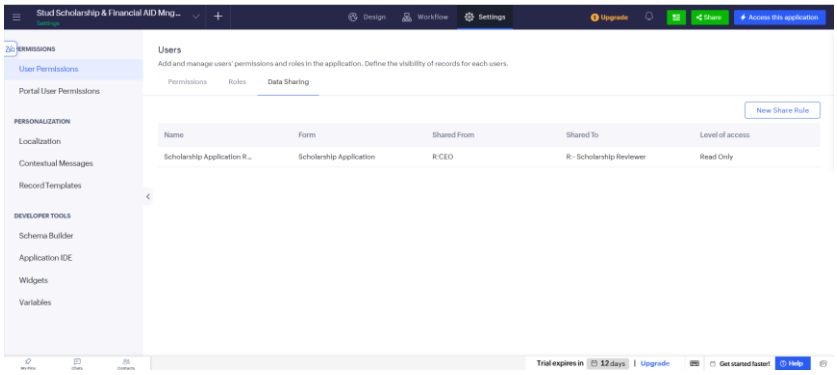


Figure 5.8.1b: Data Sharing.

5.9 AI and Analytics Features

The Scholarship Application Management System stores scholarship application details, including application dates, application status, award amounts, and eligibility status. These data fields can support basic reporting and analysis of scholarship applications.

The application data can be reviewed to monitor the number of applications received, track application statuses, identify eligible applicants, and examine scholarship award

amounts. These insights can assist administrators and scholarship reviewers in monitoring the application process and making informed decisions.

No separate AI-based prediction, recommendation, or automated decision-making feature was implemented as part of the current configuration. The system's current analytical value is based on the application data captured through the form and the reporting capabilities available in Zoho Creator.

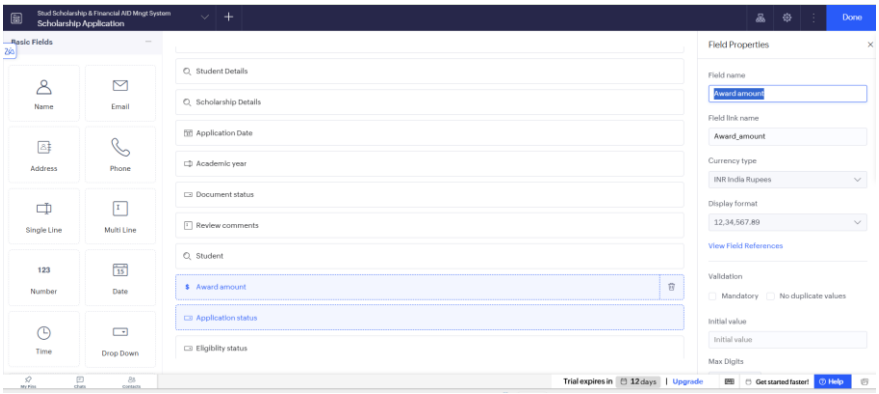


Figure 5.9: Scholarship Application Records View

5.10 Pro-Code Extension

Zoho Creator supports pro-code customization through Deluge scripting, which enables developers to implement custom validation rules and automate application processes.

In this project, a Deluge validation workflow was configured for the Scholarship Application form to validate submitted data. An email notification workflow was also configured to notify users after successful submission. In addition, Zoho Creator's integration action builder was used to send application data to Zoho CRM.

These features combine low-code configuration with custom scripting to support application validation, notifications, and CRM integration. The current implementation focuses on the requirements of the scholarship application process and can be extended in the future with additional scripts for advanced eligibility checks, automated status updates, and reporting.

5.11 Mobile and Portal Access

Zoho Creator supports mobile access to applications, allowing users to access forms and records through compatible mobile devices. This capability can help students

submit scholarship applications and enable administrators to review application information when required.

The Scholarship Application Management System was developed using Zoho Creator's form and workflow features. Mobile access and a dedicated student portal were not separately configured or tested as part of the current implementation. These features may be considered future enhancements to improve accessibility and provide students with a dedicated interface for tracking application status.

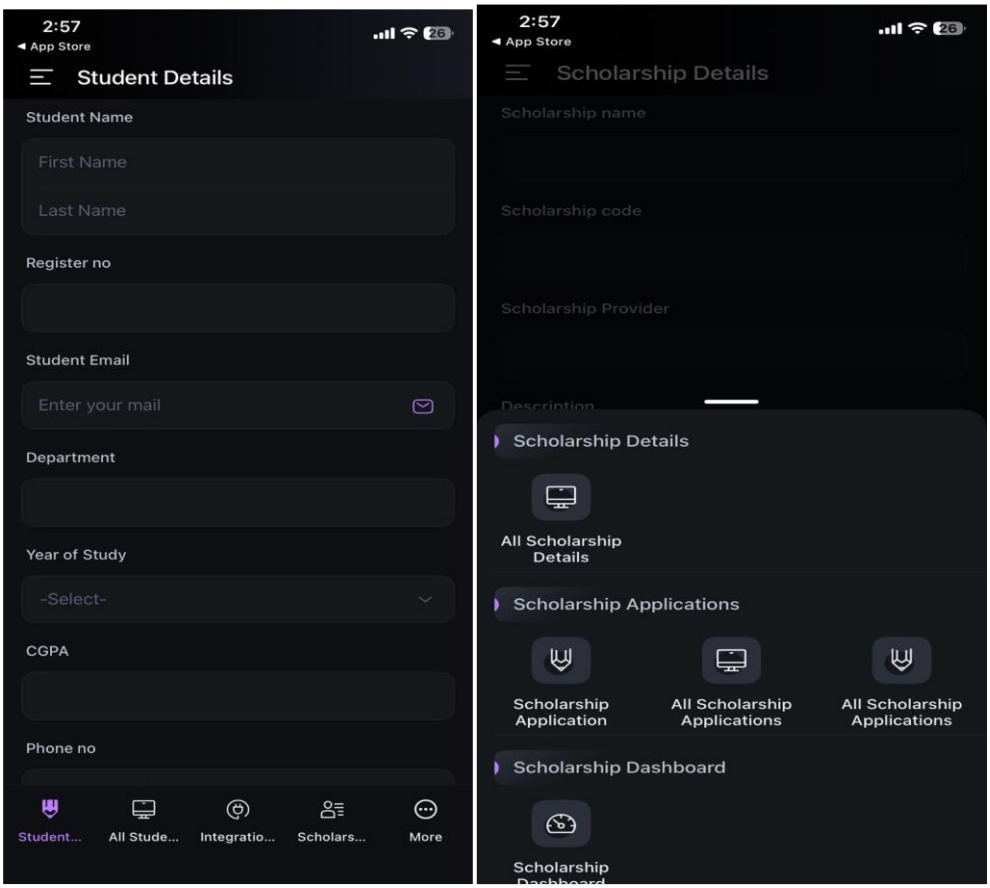


Figure 5.10: Scholarship Application Management System – Mobile Access

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing is an important phase in the development of the Stud Scholarship & Financial Aid Management System. The main objective of testing is to verify that the application form, validation workflow, email notification, and Zoho CRM integration work as expected.

The system was tested using the following approaches:

Functional Testing: Verifies that the scholarship application form accepts and stores application details.

Validation Testing: Checks whether the application form validates the information entered by the user through the Deluge validation script.

Integration Testing: Verifies that the scholarship application workflow integrates with Zoho CRM and creates a corresponding record.

Notification Testing: Checks whether an email notification is received after a successful form submission.

User Access Testing: Reviews the configured user roles and permission profiles to support appropriate access to the application.

6.2 Test Cases

Table 6.1: Test Cases

Test Case ID	Test Scenario	Expected Result	Status
TC-01	Open the scholarship application form	The form opens successfully	Pass
TC-02	Enter applicant details	Applicant information can be entered	Pass
TC-03	Check mandatory-field validation	Required fields are configured for validation	Pass

Test Case ID	Test Scenario	Expected Result	Status
TC-04	Enter personal information	Personal details can be captured	Pass
TC-05	Enter academic information	Academic details can be captured	Pass
TC-06	Enter scholarship information	Scholarship details can be captured	Pass
TC-07	Submit a scholarship application	The form supports application submission	Pass
TC-08	Store application details	Application records can be captured	Pass
TC-09	Check the validation workflow	The Deluge validation workflow is configured	Pass
TC-10	Review incomplete application handling	Validation logic is configured to handle application input	Pass
TC-11	Check email notification	An email notification is received after testing	Pass
TC-12	Check Zoho CRM integration	A test record is created successfully in Zoho CRM	Pass
TC-13	Review CRM field mapping	CRM fields are configured through the integration interface	Pass
TC-14	Review user access configuration	User roles and permission profiles are configured	Pass
TC-15	Access the application on a mobile device	The application is accessible on a mobile device	Pass

The test cases cover the major functional areas of the application. Email notification and Zoho CRM record creation were confirmed during testing. Other entries describe the implemented functionality and configuration.

6.3 Debugging Deluge Scripts

Deluge scripting is used to implement validation logic for scholarship application submissions. The validation workflow is associated with the form submission process and is intended to check applicant information before processing the application.

During development, the validation script and its workflow configuration were reviewed to ensure that the script was associated with the appropriate form event.

The Zoho CRM integration was configured using the `scholarship_crm_connection` connection. A test application was submitted successfully, and a corresponding record was created in Zoho CRM.

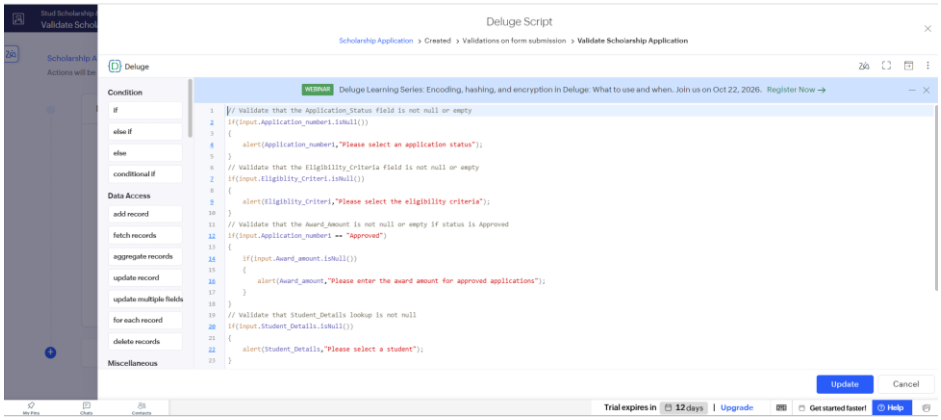


Figure 6.1: Deluge Debug Output

6.4 Defect Log

The defect log provides a record of issues identified during the development and testing of the system.

Table 6.2: Defect Log

Defect ID	Description	Severity	Status
D-01	Validation workflow configuration	Medium	Configured
D-02	Zoho CRM integration and field mapping	Medium	Integration configured
D-03	User roles and permission profiles	Medium	Configured

6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
User 1	Scholarship Applicant	The application form is easy to understand and enter details.	Reviewed the form layout and field arrangement.
User 2	Scholarship Reviewer	The application information and email notification help with application processing.	Reviewed the notification workflow and application management configuration.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The Stud Scholarship & Financial AID Management System was deployed using the multi-stage environment feature available in Zoho Creator. This feature supports managing application versions across Development, Stage, and Production environments.

The application has a published version, Version 1.0, in both the Stage and Production environments. The Version History section contains a record titled **Default Package 1.0**, with the description "Package published by default." The version type is Major, and the application version is recorded as V1.0.

The Development environment was also reviewed. The environment initially displayed a development app creation status and subsequently showed "No changes available." This indicates that no new changes were available for publishing at the time of inspection.

The multi-stage deployment approach helps separate application changes from the production version. New changes can be reviewed in the development and staging stages before they are published to production. Version History can be used to review previously published packages and their details.

A version history record provides a reference for tracking releases. Before publishing future changes, the application should be checked to ensure that its forms, workflows, permissions, and integrations continue to function correctly.

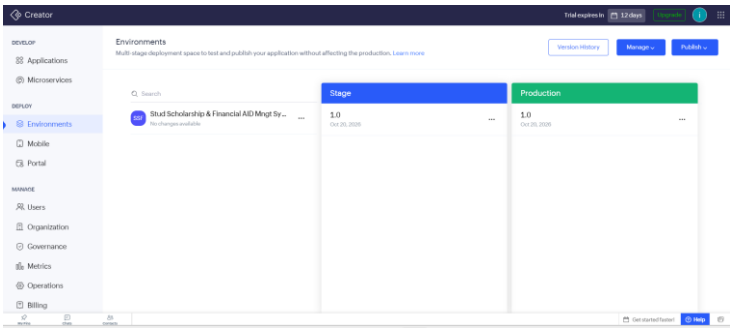


Figure 7.1: Environments and Publish to Production

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases executed in Development / Stage	✓
2	Sample/test data removed or isolated	✓
3	Roles, permissions and users configured in Production	✓
4	Connections authorised in Production	✓
5	Schedules and workflows enabled	✓
6	Application published and URL verified	✓
7	Version / release notes recorded	✓

7.2 Performance Optimisation

Performance optimisation techniques were applied to improve the efficiency of the Stud Scholarship & Financial AID Mngt System. The Scholarship Application Details report was optimised using a filter criterion on the Application Status field to display applications with the status “Under Review”. This reduces unnecessary records in the report view and helps users locate relevant scholarship applications efficiently.

Criteria-based data retrieval, avoiding repeated fetch operations inside loops, selecting only necessary report columns, and using bulk operations where appropriate are important optimisation practices for reducing redundant processing and improving application responsiveness.

The implemented report filter improves the relevance of displayed data and supports efficient application monitoring.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Replaced fetch inside loop with single criteria fetch	DS-03	Repeated fetch operations	Single criteria-based fetch	Reduces redundant database fetches
Filtered report by default criteria	Scholarship Application Details	All application records displayed	Only records matching Application Status = Under Review displayed	Reduces irrelevant records displayed

7.3 Security Features

Security features are important for protecting student information and scholarship application data in the Stud Scholarship & Financial AID Mngt System. Access control, record-level permissions, field-level security, secure authentication, and protected connections help prevent unauthorised access and safeguard sensitive information.

The application security approach focuses on restricting access to relevant information, protecting confidential student data, and ensuring that application records are accessed and managed appropriately. Regular review of access permissions and secure administration practices support the confidentiality and integrity of the application.

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Record-level permission criteria	Scholarship Application records	Unauthorised data access
Field-level security	Student Details form	Exposure of sensitive data
Secure connection / OAuth	External integrations, where configured	Credential leakage

Security Control	Where Applied	Threat Mitigated
Authentication and access permissions	Zoho Creator application	Unauthorised application access

Security features were configured in the Stud Scholarship & Financial AID Mngt System to protect student and scholarship application data. Read and Write permissions were configured through User Permissions, and the Scholarship Viewer role was established to control application access. Data Sharing settings were configured to manage access to records. Security policies were also configured through the Governance panel to strengthen application security and support controlled access to information.

7.4 Governance and Administration

Governance and administration practices were considered to maintain the security, reliability, and maintainability of the Stud Scholarship & Financial AID Mngt System. Security policies were configured through the Governance panel to strengthen access control and improve organizational security.

Role-based permissions, the Scholarship Viewer role, and Data Sharing settings support controlled access to application data. Proper naming conventions and regular review of application configurations help maintain consistency and simplify future maintenance. Application data can be exported periodically for backup and record-keeping purposes. Administrative responsibilities include reviewing user access, maintaining application configurations, monitoring changes, and ensuring that student and scholarship information is managed securely.

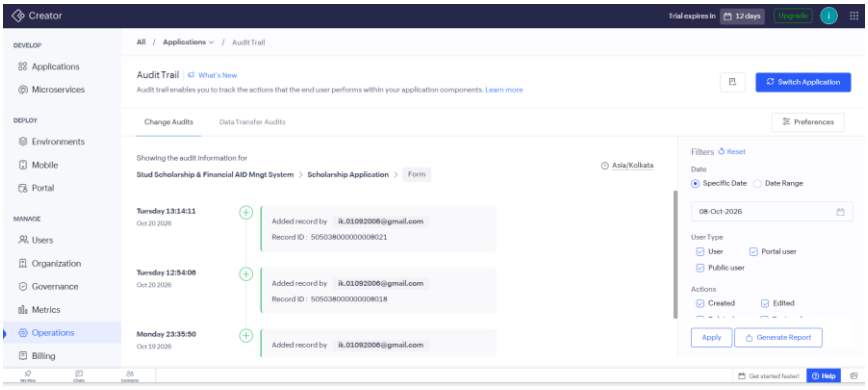


Figure 7.2: Audit Trail / Administration Settings

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The working scenario begins with entering student information into the Student Details form. Next, scholarship application information is entered through the Scholarship Application form by selecting the relevant student and scholarship details. Finally, the Scholarship Application Details report is used to view and review the available application information. The application status filter helps focus on applications with a selected status, such as Under Review. This walkthrough demonstrates the main data-entry and report-viewing stages of the Stud Scholarship & Financial AID Mngt System.

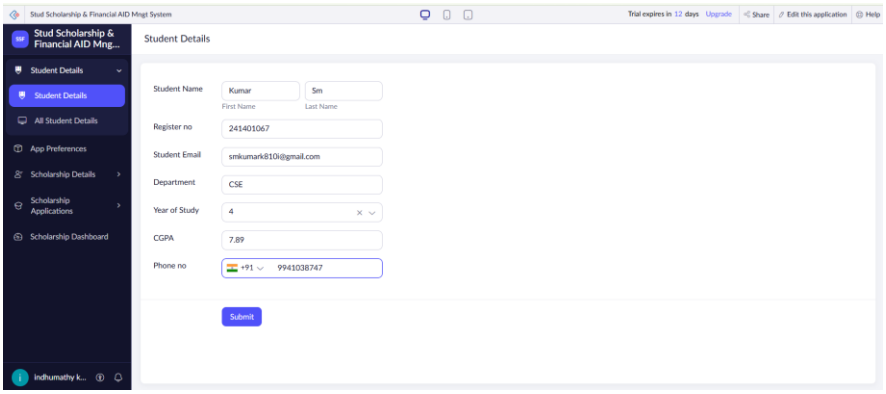


Figure 8.1: Scenario Step 1 – This screenshot shows the Student Details form of the Stud Scholarship & Financial AID Mngt System. It demonstrates how student information is entered into the application form as the first step in the scholarship management process.

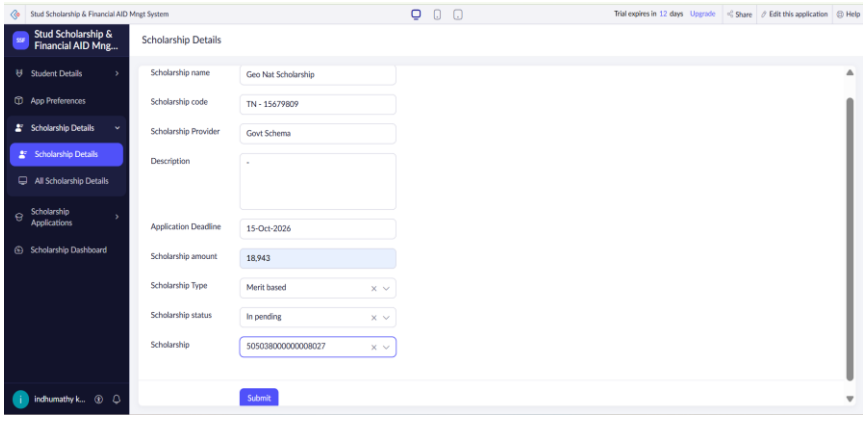


Figure 8.2: Scenario Step 2 – Scholarship Application Entry. This screenshot shows the Scholarship Application form used to enter scholarship application details and associate the application with the relevant student and scholarship information.

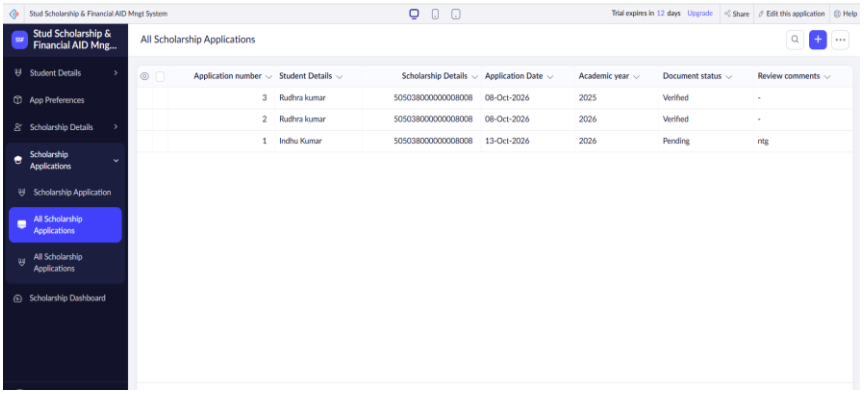


Figure 8.3: Scenario Step 3 – Scholarship Application Report. This screenshot shows the Scholarship Application Details report, which provides an organised view of scholarship application information and supports reviewing records using the available application status filter.

8.2 Achievement of Objectives

The objectives of the Stud Scholarship & Financial AID Mngt System were addressed through the forms, lookup relationships, reports, filtering options, and access-control settings configured in Zoho Creator. The following table maps the project objectives to their implementation and outcomes.

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
To maintain student information systematically	Student Details form (Section 5.x)	Provides a structured form for entering student information
To maintain scholarship information	Scholarship Details form (Section 5.x)	Organises scholarship-related information
To manage scholarship applications	Scholarship Application form and lookup fields (Section 5.x)	Supports entering application details and associating them with related information
To view and review application information	Scholarship Application Details report and status filter (Section 5.x)	Provides an organised report view with status-based filtering

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
To control access to application information	User Permissions, Scholarship Viewer role, and Data Sharing settings (Section 5.x)	Supports controlled access to application records

The project provides a structured foundation for managing student details, scholarship information, and scholarship applications. The configured forms, reports, filters, and access settings address the main objectives of organising information and supporting application review.

8.3 Comparison with the Existing System

The existing system may rely on manual records, separate documents, or spreadsheets to maintain student and scholarship information. The proposed application uses Zoho Creator forms, lookup fields, reports, and filters to organise related information in one application.

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Time to process one request	May require manually searching and updating records	Uses dedicated forms and organised report views	Supports a more structured process and reduces unnecessary searching
Manual steps	Information may be entered and maintained in separate records	Information is entered through dedicated application forms	Organises data entry in a consistent manner
Report generation	Information may need to be compiled manually	Application report displays organised application information	Makes reviewing available information more convenient

The proposed application organises scholarship-related data and provides dedicated forms and reports for information management. These features improve the structure and accessibility of records compared with a process that relies entirely on separate manual records. Actual time savings depend on usage and the amount of data being processed.

8.4 Limitations of the Low-Code Approach

Low-code platforms simplify application development by providing visual builders, predefined components, and configuration options. However, some limitations must be considered while developing applications using Zoho Creator.

The first limitation is UI customisation. The form and report design depends on the layout options and components provided by the platform. This can be handled by using the available field properties, form layouts, and report settings effectively.

The second limitation concerns complex business logic. Basic requirements can often be implemented through configuration, while advanced logic may require Deluge scripting. In this project, the available forms, lookup relationships, reports, and filters were used to organise the main application functions without introducing unnecessary complexity.

Third, integrations and API usage may be subject to platform permissions, usage limits, and service availability. Applications requiring external integrations must be designed according to the supported APIs and applicable limits.

Finally, vendor lock-in is a concern because the application depends on Zoho Creator's platform and services. Maintaining documentation of forms, fields, relationships, reports, and application settings helps support future maintenance and migration planning.

Overall, the project demonstrates how built-in low-code features can be used to develop a structured application while recognising that advanced customisation, complex logic, external integrations, and platform dependency may require additional planning.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

The development of the Stud Scholarship & Financial AID Mngt System illustrates the differences between no-code, low-code, and pro-code development approaches.

No-code configuration: Creating forms such as Student Details, Scholarship Details, and Scholarship Application, configuring fields, establishing lookup relationships

through the visual interface, and setting up reports and filters can be performed using the platform's built-in configuration tools.

Low-code development: Deluge scripting can be used when additional validation, automated notifications, custom business rules, or more advanced workflows are required. Such scripting provides flexibility beyond basic visual configuration. The need for scripting depends on the specific requirements of the application.

Pro-code development: A fully custom-coded solution may be considered when requirements exceed the platform's capabilities, such as highly customised interfaces, specialised algorithms, complex external integrations, or greater control over application architecture.

This project demonstrates that low-code development offers a balance between ease of configuration and the ability to extend functionality when needed. A move towards pro-code development would be appropriate if future requirements demand greater customisation, complex processing, or independence from a specific low-code platform. The choice should depend on application complexity, maintenance needs, available skills, and long-term goals.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Stud Scholarship & Financial AID Mngt System was developed using Zoho Creator, a low-code application development platform. The application includes Student Details, Scholarship Details, and Scholarship Application forms, with lookup relationships to connect related information. The Scholarship Application Details report and application status filter support organised viewing and review of scholarship application information.

The project also involved configuring user permissions, the Scholarship Viewer role, data sharing settings, and security policies to support controlled access to application data. Overall, the project demonstrates how low-code tools can be used to develop a structured application for scholarship information management while gaining practical experience in form configuration, data relationships, reporting, filtering, and application security.

9.2 Future Enhancements

- **Predictive Analytics using Zia:** Integrate Zoho Zia or suitable analytics tools to analyse application trends and provide insights that assist scholarship administrators in decision-making.

- **Automated Notifications and Payment Integration:** Introduce automated email or SMS notifications for application status updates. If scholarship disbursement management is required, a secure payment or fund-disbursement integration could also be considered.

- **Catalyst-Based Microservice for Advanced Processing:** Integrate Zoho Catalyst or another suitable backend service for complex processing, advanced integrations, and operations that exceed the application's built-in capabilities.

9.3 Learning Outcomes and Career Relevance

This project provided practical experience in developing a low-code application using Zoho Creator. The development process improved understanding of form creation, field configuration, lookup relationships, report generation, filtering, user permissions, and application security. It also demonstrated the importance of organising data, maintaining appropriate access controls, and presenting application information clearly.

The skills gained through this project are relevant to several career pathways. A **Citizen Developer** uses low-code or no-code tools to create business applications with limited programming. A **Low-Code Developer** builds and extends applications using visual development tools and scripting. A **Solutions Consultant** analyses business requirements and designs suitable technology solutions. An **Automation Engineer** develops workflows and integrations to reduce repetitive tasks and improve process consistency.

The project also highlighted the importance of understanding platform limitations, testing application behaviour, and identifying opportunities for future enhancements. As low-code platforms continue to evolve, the combination of application development knowledge, problem-solving skills, and an understanding of business processes can support further learning and career development in software and automation-related fields.

REFERENCES

[1] A. C. Bock and U. Frank, “Low-Code Platform,” *Business & Information Systems Engineering*, vol. 63, pp. 733–740, 2021, doi: 10.1007/s12599-021-00726-8.

[2] R. Waszkowski, “Low-code platform for automating business processes in manufacturing,” *IFAC-PapersOnLine*, vol. 52, no. 10, pp. 376–381, 2019, doi: 10.1016/j.ifacol.2019.10.060.

[3] Zoho Corporation, “Understand Application,” *Zoho Creator Help*. [Online]. Available: https://help.zoho.com/portal/en/kb/creator/developer-guide/creating-applications/understand-applications/articles/understand-application. [Accessed: Oct. 20, 2026].

[4] Zoho Corporation, “Creating and managing form workflows,” *Zoho Creator Help*. [Online]. Available: https://help.zoho.com/portal/en/kb/creator/developer-guide/workflows/create-and-manage-form-workflows/articles/create-new-form-workflow. [Accessed: Oct. 20, 2026].

[5] Zoho Corporation, “Understanding application creation with AI using Zia App Builder,” *Zoho Creator Help*. [Online]. Available: https://help.zoho.com/portal/en/kb/creator/developer-guide/creating-applications/create-applications/articles/creating-applications-with-zia-app-builder. [Accessed: Oct. 20, 2026].

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

> ***Guidance:*** *Mark each feature as implemented or not, state where it lives in the application (form/report/page/workflow name) and the report section and figure that evidence it. Examiners verify this table against the live application.*

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	[Y / N]	[ ]	5.1	[ ]
II	Forms with ≥ 8 distinct field types	[Y / N]	[ ]	5.2	[ ]
II	Lookup field (relationship between forms)	[Y / N]	[ ]	4.2, 5.2	[ ]
II	Subform (parent–child data)	[Y / N]	[ ]	4.2, 5.2	[ ]
II	Field validations / layout rules	[Y / N]	[ ]	5.2	[ ]
II	List report with filter, grouping, custom action	[Y / N]	[ ]	5.3	[ ]
II	Kanban / Calendar / Timeline / Map report	[Y / N]	[ ]	5.3	[ ]
II	Pivot table / chart report	[Y / N]	[ ]	5.3	[ ]
II	Page / dashboard built with UI Builder	[Y / N]	[ ]	5.4	[ ]
II	Deluge – form events (validate / success / field actions)	[Y / N]	[ ]	5.5	[ ]
II	Deluge – functions, fetch, loops, collections	[Y / N]	[ ]	5.5	[ ]
II	Email / in-app notifications	[Y / N]	[ ]	5.5, 5.6	[ ]
II	Form workflow	[Y / N]	[ ]	5.6	[ ]
II	Scheduled workflow	[Y / N]	[ ]	5.6	[ ]
II	Approval process	[Y / N]	[ ]	5.6	[ ]

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
II	Blueprint	[Y / N]	[ ]	5.6	[ ]
II	Batch / report workflow	[Y / N]	[ ]	5.6	[ ]
II	Testing & debugging evidence	[Y / N]	[ ]	6	[ ]
III	Integration with a Zoho service	[Y / N]	[ ]	5.7	[ ]
III	Connection / REST API call (invokeurl)	[Y / N]	[ ]	5.7	[ ]
III	Webhook or Creator REST API	[Y / N]	[ ]	5.7	[ ]
III	Users, roles and permission profiles	[Y / N]	[ ]	5.8	[ ]
III	Record-level access criteria / customer portal	[Y / N]	[ ]	5.8	[ ]
III	Zia / AI feature	[Y / N]	[ ]	5.9	[ ]
III	Analytics (charts / Zoho Analytics)	[Y / N]	[ ]	5.9	[ ]
IV	Performance optimisation applied & measured	[Y / N]	[ ]	7.2	[ ]
IV	Deployment across environments	[Y / N]	[ ]	7.1	[ ]
IV	Security controls (encryption, OAuth, permissions)	[Y / N]	[ ]	7.3	[ ]
IV	Governance (audit trail, administration)	[Y / N]	[ ]	7.4	[ ]
V	Pro-code extension (Catalyst / widgets / custom functions)	[Y / N]	[ ]	5.10	[ ]
V	Low-code limitations & spectrum analysis	[Y / N]	[ ]	8.4, 8.5	[ ]
—	Mobile app / portal access	[Y / N]	[ ]	5.11	[ ]

Total features implemented: [ ] / 32

APPENDIX B – COURSE OUTCOME MAPPING

> ***Guidance:*** *Show where each Course Outcome of CB23F35 is evidenced in this report and application. Self-assess the attainment level (1 – Low, 2 – Medium, 3 – High); the examiner may revise it.*

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	[ ]	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	[ ]	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	[ ]	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	[ ]	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	[ ]	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

> ***Guidance:*** *Paste every Deluge script in full, grouped by form/function, with a header comment giving the Script ID (from Table 5.x), location/trigger and purpose. Use Courier New 10 pt in the shaded code box. Do not paste screenshots of code.*

C.1 [Form / Function Name]

```
// Script ID : [DS-01]
// Location  : [Form → Event / Function name]
// Purpose   : [ ]

[paste code here]
```

C.2 [Form / Function Name]

```
// Script ID : [DS-02]
// Location  : [ ]
// Purpose   : [ ]

[paste code here]
```

APPENDIX D – USER MANUAL

> ***Guidance:*** *Write short, step-by-step instructions for each role so that a new user can operate the application without help. Include login, key tasks and where to find reports.*

D.1 Accessing the Application

[Steps to log in on web and mobile.]

D.2 Instructions for [Role 1]

- [Step 1]
- [Step 2]

D.3 Instructions for [Role 2]

- [Step 1]
- [Step 2]

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

> ***Guidance:*** *Suggested distribution for the End Semester Examination (Individual Project). The course coordinator may revise the weightage to align with Controller of Examinations norms before release to students.*

Student Name	[ ]	Register No.	[ ]
Project Title	[ ]	Date	[ ]

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	[ ]	Register No.	[ ]
Department / Section	[ ]	Date of Viva	[ ]
Application URL Verified	Yes / No	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER

Name: _____

EXTERNAL EXAMINER

Name: _____